Alex Lewandowski

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Skills

Robotics & Controls: PID, feedforward, odometry, sensor fusion, kinematics, state machines, ROS2
CAD & Mechanical Design: Fusion 360, SolidWorks, 3D printing
Software: Python, C++, Java, Git
Embedded Systems: PROS, OkapiLib, **ESP32**, **Teensy 4.1**, I2C
Machine Learning & CV: OpenCV, NEAT (neuroevolution)
Manufacturing & Prototyping: Iterative prototyping, load testing, mechanical fastening design

Experience

Humanoid Perception Team Member Sep 2026 - Present
WATonomous, University of Waterloo

- Built humanoid SLAM backend replay tooling in **GTSAM/iSAM2**, integrating VO keyframes and loop closures into pose graphs and exporting optimized trajectories.
- Validated **Huber-robust optimization** with deliberately corrupted loop closures; compared frontend and backend trajectories using **EVO APE** to measure the effect of bad constraints.
- Developed **RTAB-Map/EVO** benchmarking on **TUM RGB-D** and **RealSense D455** bags using APE/RPE, trajectory plots, and loop-closure error without ground truth.
- Built Dockerized ROS 2 tooling for bag conversion, topic/frame configuration, trajectory export, and D455 recording; diagnosed TF and timestamp issues.
- Implemented ROS 2 navigation in simulation with **LiDAR costmaps**, **A-star planning**, and **Pure Pursuit**.

Head Programmer, Designer & Strategist Jun 2022 - Jun 2026
VEX Robotics Team 5225A, The PiLons

- Won the **2025 Ontario Excellence Award** and **VEX Worlds Design Award**; placed **4th of 10,000+ teams worldwide** in **Autonomous Skills**. Recognized for an organized engineering design process and notebook.
- Won the **Think Award** at 2024 and 2026 VEX Worlds for clean, documented autonomous code and clearly explained programming strategy.
- Led **Fusion 360** mechanical design, redesigning the robot between events to adapt competition strategy.
- Engineered a **power take-off (PTO)** that repurposed **X-drive motors** for hanging, an approach unique among X-drive teams that season.
- Fused gyroscope, encoder, and distance-sensor data for **odometry accurate to ±0.02 m**.
- Applied kinematics and **inverse kinematics** for precise robotic-arm positioning and game-element manipulation.
- Designed closed-loop **PID and feedforward** velocity controllers for real-time autonomous motion.
- Developed state machines coordinating subsystems during autonomous and driver-controlled operation.

Projects

World Robot Olympiad Robot ***ESP32, Teensy 4.1****, Sensor Integration, CAD* Oct 2025 - May 2026

- Qualified for the **California Invitational Open**; fielded the only robot at Nationals using pneumatics and **custom-designed electronics**.
- Split compute between **ESP32** sensing/localization and **Teensy 4.1** motor control, integrating a **9-axis IMU**, dual color sensors, and camera for field-element detection.

Differential Swerve Drivetrain ***Fusion 360****, Mechanical Design, Load Testing* Sep 2025 - Dec 2025

- Designed **differential modules** using two motor speeds: their sum drives the wheel; their difference steers.
- Reinforced the chassis with triangle braces and a **steel-ball thrust bearing**.

Education

University of Waterloo - B.A.Sc., Mechatronics Engineering Sep 2026 - Jun 2031
President's Scholarship of Distinction